MOHAMAD AYHAM AYAN

AI / Machine Learning Engineer | Document AI · Computer Vision · LLMs & Agents

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PROFESSIONAL SUMMARY

AI/ML Engineer with a BSc in Information Technology Engineering, building **production document-AI, computer-vision and LLM systems** in an applied AI pod. Covers the full loop: spec-driven VLM extraction, golden-set annotation, **LoRA/QLoRA fine-tuning** and **vLLM serving** on an **NVIDIA DGX Spark**, and **per-field evaluation** tracked in W&B. Ships models behind tested, containerised **FastAPI services**, with Celery or **Temporal for long-running work**. Also builds **multi-agent LLM workflows** and **local RAG** for Arabic and English.

EXPERIENCE

AI/ML Engineer — Applied AI Pod, Metamas Technologies · Aleppo, Syria

Jul 2026 – Present

Document-AI platform for Arabic work orders

FastAPI · Celery · Redis · PostgreSQL · Gemini · Qwen3.5-VL · Unsloth · vLLM · W&B · Docker

- Built a **spec-driven engine** that turns image-only Arabic PDFs into typed, section-structured JSON. One declarative catalog of **123 fields across 31 document types** generates every prompt and JSON schema, so a new document type needs no engine change.
- Held cost to **5–6 VLM calls per 5-page document** with **CPU-only page classification** (OpenCV template matching, HSV masks) and **cheapest-first field routing**. Added deterministic normalisation, cross-field checks and typed issues that send uncertain documents to human review.
- Closed the training loop: an annotation UI replays recorded model calls to build a golden set of **183 reviewed documents** and exports LoRA datasets. Fine-tuned **Qwen3.5-VL (4B/9B) with LoRA/QLoRA** through a resumable GPU trainer service on DGX Spark, and **served adapters with vLLM** behind an OpenAI-compatible API.
- Built per-field evaluation against the golden set (DocScore, straight-through-processing rate, critical-field pass rate) mirrored to **W&B**, plus a prompt playground that separates prompt failures from model failures. The stack runs as 8 Docker services, 52 API paths and **1,130 tests** that need no GPU or API key.

Real-time PPE compliance monitoring

YOLO26 · InsightFace · FastAPI · Celery · Redis · WebSocket · React · Docker

- Built a worksite PPE monitor around a crop-then-detect pipeline. A YOLO26 pose model finds each worker, then a **fine-tuned YOLO26 PPE detector** checks the crop. A **tri-state state machine** (compliant, violation, indeterminate) with a grace window confirms violations.
- Serves both **uploaded video** (async Celery jobs) and **live RTSP or browser-camera streams** over WebSocket. Only GPU workers hold models, and a latest-wins frame buffer keeps live latency flat.
- Identifies violators with **InsightFace (SCRFD + ArcFace)** against an enrolled gallery, aggregating quality-weighted embeddings per track to keep identities stable. Frame, person and face evidence is stored for each violation.
- Fine-tuned **YOLO26s on DGX Spark** (MuSGD, cosine LR, mosaic/mixup/copy-paste augmentation, 150 epochs) with checkpoint auto-resume and an out-of-memory batch fallback, then exported NMS-free ONNX.

VLM inspection-rule extraction from engineering standards

Temporal · PostgreSQL · Alembic · OpenRouter · PyMuPDF · React

- Built a pipeline that turns construction-standard PDFs into **atomic, citation-backed inspection rules**, each phrased as one yes/no question a VLM can check against a site photo.
- Ran multi-pass extraction (index, batched extraction, duplicate sweep, verification) as **Temporal workflows**, so a crashed worker resumes without re-paying for finished batches. Every cited quote is **string-matched** against its source page.
- Added versioned prompts and extraction profiles, cost estimates from the live OpenRouter catalogue, and **LLM-as-judge** comparison across runs.

Camera calibration & tilt-compliance web app

FastAPI · React / TypeScript · GeoCalib · OWLv2 · Docker · GitLab CI

- Wrapped **single-image camera calibration (GeoCalib)** in a web app that finds an object by name with **OWLv2 zero-shot detection** and checks its tilt against a tolerance. Inference is split from rendering so overlays redraw instantly. Fixed a shared-model race with **per-model locks**, and runs non-root CPU/GPU Docker with pytest and Vitest in CI.

SAM 3 auto-labelling tool

SAM 3 · Transformers · PyQt6 · FastAPI · Celery · React · Docker

- Built a text-prompted labeller that turns a folder or ZIP of images into a ready **YOLO dataset** with **SAM 3**. It ships as a PyQt6 desktop app and as a Dockerised web app with a GPU Celery worker.

AI & Machine Learning Intern — Midad Organization, with Tamkin for IT *May 2025 – Present*

- Delivered a capstone ML system forecasting internal displacement (IDP) flows in Syria from ACAPS and HDX data, **awarded 100/100**. Used **SHAP** to make its predictions auditable.

Additional Experience

- **Freelance E-Commerce & Digital Sales** (2022–2023) — Consumer electronics · social-media marketing · lead generation · B2C sales & negotiation · customer support · order fulfilment.
- Private Mathematics Tutor (freelance, 2022–2024) · Pharmacy Assistant, Al Hatabi Pharmacy (2018)

TECHNICAL PROJECTS

Multi-Agent Guest Operations Copilot (MVP) *2026*

Google ADK · Temporal · FastAPI · PostgreSQL · LiteLLM · React

- **Five ADK agents** (orchestrator, reservation, policy, response, review/safety) handle each guest message inside a **durable Temporal workflow** that waits for human approval. Facts come only from tool results, never from model text.
- Added a **deterministic decision gate** that can only make outcomes stricter. In testing it escalated a gas-leak report that a real LLM reviewer had approved. Added **offline agent evals**, idempotent ingestion and switchable providers (Gemini, Cohere, OpenRouter).

Chat with Your Data — Fully Local Text-to-SQL RAG *2026*

Qwen (local LLM) · LlamaIndex · Qdrant · Oracle 19c

- Built an Arabic/English natural-language interface to an Oracle 19c database with **schema-aware retrieval**. It runs **fully on-premise**, so no data leaves the server.

IDP Flow Forecasting · Face Generation with GANs *2025*

scikit-learn · Random Forest · SHAP · TensorFlow · CNN · GAN

- Built **Random Forest forecasting** on sparse conflict-zone data. Built a **UTKFace GAN** and fixed mode collapse with label smoothing.

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, C#
LLMs, VLMs & Agents	Qwen3.5-VL, Gemini, LoRA/QLoRA (Unsloth), vLLM, structured output and JSON schemas, Google ADK, LiteLLM, OpenRouter, LLM-as-judge
RAG	LlamaIndex, Qdrant, Text-to-SQL, local LLM deployment
Computer Vision	YOLO (v8 / 11 / 26), pose estimation, InsightFace (SCRFD, ArcFace), SAM 3, OWLv2 zero-shot detection, GeoCalib, OpenCV, ONNX export
ML & Evaluation	PyTorch, TensorFlow, scikit-learn, golden-set evaluation, W&B, SHAP, cross-validation
Backend & Orchestration	FastAPI, WebSocket, Celery, Redis, Temporal, PostgreSQL, SQLAlchemy, Alembic, Pydantic, PyMuPDF
Infra & Tools	Docker / Compose, NVIDIA DGX Spark (GB10), GitLab CI, pytest, React, PyQt6, Git, Jupyter

EDUCATION

BSc in Information Technology Engineering — University of Aleppo *Nov 2025*

Baccalaureate Certificate — Grade: 95.83% *Jun 2020*

Ninth Grade Certificate — Grade: 92.25% *Jul 2017*

TRAINING & CERTIFICATIONS

- Applied AI & Machine Learning Programme — Tamkin for IT / Midad · Neural Networks & Deep Learning Workshop
- TOT, HR Management, Public Relations, Marketing, Executive Secretariat — Mastry, Aleppo · Regulatory Team Member — BlueBits, University of Aleppo

LANGUAGES

Arabic — Native · **English** — B2, strong technical reading and writing · **German, French** — Basic